

Assignment02 - Problem4

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Problem 4.

```
library(reticulate)
```

```
Warning: package 'reticulate' was built under R version 4.4.3
```

```
py_config() # r 환경에서의 python 경로
```

```
python:      C:/Users/user/OneDrive/문서/.virtualenvs/r-
reticulate/Scripts/python.exe
libpython:    C:/ProgramData/anaconda3/python311.dll
pythonhome:   C:/Users/user/OneDrive/문서/.virtualenvs/r-reticulate
version:      3.11.3 | packaged by Anaconda, Inc. | (main, Apr 19
2023, 23:46:34) [MSC v.1916 64 bit (AMD64)]
Architecture: 64bit
numpy:        C:/Users/user/OneDrive/문서/.virtualenvs/r-
reticulate/Lib/site-packages/numpy
numpy_version: 2.2.4
```

```
NOTE: Python version was forced by VIRTUAL_ENV
```

```
py_install(c('pandas', 'scikit-learn')) # 필요한 python 패키지 설치
```

```
Using virtual environment "C:/Users/user/OneDrive/문서/.virtualenvs/r-
reticulate" ...
```

```
+ "C:/Users/user/OneDrive/문서/.virtualenvs/r-
reticulate/Scripts/python.exe" -m pip install --upgrade --no-user
pandas scikit-learn
```

(a).

```
library(dplyr)
```

```
Warning: package 'dplyr' was built under R version 4.4.3
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':
```

```
filter, lag
```

```
The following objects are masked from 'package:base':
```

```
intersect, setdiff, setequal, union
```

```
set.seed(1)
```

```
data(iris)
```

```

n <- nrow(iris)
index <- sample(n, size = 0.7*n)
train <- iris[index, ]; test <- iris[-index, ]

write.csv(train, 'data/train.csv', row.names = FALSE)
write.csv(test, 'data/test.csv', row.names = FALSE)

```

(b).

```

import pandas as pd
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import confusion_matrix, accuracy_score

train = pd.read_csv('data/train.csv')
test = pd.read_csv('data/test.csv')

X_train = train.drop('Species', axis = 1)
y_train = train['Species']
X_test = test.drop('Species', axis = 1)
y_test = test['Species']

rf = RandomForestClassifier(random_state = 1)
rf.fit(X_train, y_train)

RandomForestClassifier(random_state=1)

y_pred = rf.predict(X_test)
print(y_pred)

['setosa' 'setosa' 'setosa' 'setosa' 'setosa' 'setosa' 'setosa'
'setosa'
'setosa' 'setosa' 'setosa' 'setosa' 'setosa' 'setosa' 'setosa'
'versicolor' 'versicolor' 'versicolor' 'versicolor' 'versicolor'
'versicolor' 'versicolor' 'versicolor' 'versicolor' 'versicolor'
'versicolor' 'versicolor' 'versicolor' 'versicolor' 'versicolor'
'versicolor' 'versicolor' 'virginica' 'versicolor' 'virginica'
'virginica' 'virginica' 'virginica' 'virginica' 'virginica'
'virginica'
'virginica' 'virginica' 'versicolor' 'virginica']

confusion_mat = confusion_matrix(y_test, y_pred)
acc = accuracy_score(y_test, y_pred)

```

(c).

```

conf_mat <- py$confusion_mat
acc <- py$acc

conf_mat

```

```

      [,1] [,2] [,3]
[1,]   15    0    0
[2,]    0   17    0
[3,]    0    2   11

```

```

library(ggplot2)
library(reshape2)

```

Warning: package 'reshape2' was built under R version 4.4.3

```

cm_df <- melt(conf_mat)
colnames(cm_df) <- c('True', 'Pred', 'Count')

```

```

cm_df$True <- factor(cm_df$True, levels = sort(unique(cm_df$True)))
cm_df$Pred <- factor(cm_df$Pred, levels = sort(unique(cm_df$Pred)))

```

```

ggplot(cm_df, aes(x = Pred, y = True, size = Count, color = Count)) +
  geom_point(alpha = 0.75) +
  labs(x = 'Predicted Class', y = 'True Class',
       size = 'Count') +
  theme_minimal()

```

